

Duc-Tuan Truong

Address: *Jurong West, Singapore*
 Google Scholar: *Duc-Tuan Truong*
 Github: *github.com/ductuantruong*



Student Pass holder
 Mobile/WhatsApp: *+65 96113369*
 Email: *ductuanpy99@gmail.com*
 LinkedIn: *linkedin.com/ductuantruong*

OVERVIEW

Final-year Ph.D. student focused on speech AI, with research in deepfake speech detection, speaker recognition, and speaker age estimation. Experienced in developing new training pipelines and model architectures, leveraging state-of-the-art self-supervised speech foundation models and speech LLMs to improve downstream tasks.

EDUCATION

- **Nanyang Technological University** Singapore
Doctor of Philosophy (Ph.D.) in Computer Science; GPA: 4.83/5.0 Jan 2023 - Jan 2027 (expected)
- **Nanyang Technological University** Singapore
Master of Science in Artificial Intelligence; GPA: 4.55/5.0 Jan 2022 - Jan 2023
- **Ho Chi Minh University of Technology (HCMUT) - Vietnam National University** HCMC, Vietnam
Bachelor of Engineering in Computer Science; GPA: 8.57/10.0 Aug 2017 - Aug 2021
One of 40 selected students among 300 Computer Science students for the Talented Engineer Program.

WORK EXPERIENCE

- **American Express - GenAI Team** Singapore
Data Science Intern May 2026 - Jul 2026
 - Designed a comprehensive speech analysis pipeline utilizing deep learning and SpeechLLM-based models for speaker diarization, speaker verification, speech deepfake detection and emotion recognition.
 - Validated the pipeline on 20,000 customer calls and found considerable correlations between speech features and fraudulent transactions.
- **Hong Kong Polytechnic University** Hong Kong
Research Student Attachment Programme Feb 2025 - Jun 2025
 - Designed and evaluated novel training techniques for deepfake speech detection [[6]].
 - Implemented a proof-of-concept explainable Deepfake Speech Detection system using SpeechLLM Qwen2.5 Omni model.
- **Zalo AI** HCMC, Vietnam
Associate AI Engineer (Full-time) Aug 2021 - Jan 2022
 - Contributed to *KiKi* - the first Vietnamese voice assistant deployed on smart speakers, cars, and smartphones.
 - Parsed ASR outputs into intents and implemented intent-ranking logic for voice queries.
 - Built a streaming pipeline to update NER entities by crawling and processing data from multiple websites.
 - Analyzed & visualized ~150,000 daily user logs to diagnose failure cases and prioritize fixes.
- **Innotech** HCMC, Vietnam
AI Engineer (Full-time) Nov 2020 - Jul 2021
 - Built centralized log management and deployed AI models for streaming anomaly detection on servers using log messages.
 - Adapted reinforcement learning (Deep Q-learning, AlphaZero) to solve FreeCell.

PUBLICATIONS (* DENOTES CO-FIRST AUTHOR) (TOTAL CITATIONS: 240+, H-INDEX: 7)

- [1] **D.-T. Truong**, R. Tao, H.-T. Luong, T. Nguyen, K.A. Lee, C.E. Siong, *Temporal-Channel Modeling in Multi-head Self-Attention for Synthetic Speech Detection*, **ISCA INTERSPEECH 2024**, *GitHub stars: 60+*.
- [2] **D.-T. Truong**, R. Tao, J.Q. Yip, K.A. Lee, C.E. Siong, *Emphasized Non-Target Speaker Knowledge in Knowledge Distillation for Automatic Speaker Verification*, **IEEE ICASSP 2024**, *GitHub stars: 10+*.
- [3] T. Liu, **D.-T. Truong**, R.K. Das, K.A. Lee, H. Li, *Nes2Net: A Lightweight Nested Architecture for Foundation Model Driven Speech Anti-spoofing*, **IEEE Transactions on Information Forensics and Security 2025**, *GitHub stars: 100+*.
- [4] R. Tao, Z. Shi, Y. Jiang, **D.-T. Truong**, C.E. Siong, M. Alioto, H. Li, *Multi-Stage Face-Voice Association Learning with Keynote Speaker Diarization*, **ACM Multimedia 2024**, *GitHub stars: 22*.
- [5] **D.-T. Truong**, T. Liu, R. Tao, J. Li, K.A. Lee, C.E. Siong, *QAMO: Quality-aware Multi-centroid One-class Learning For Speech Deepfake Detection*, **ISCA INTERSPEECH 2026**, *GitHub stars: 10+*.
- [6] **D.-T. Truong**, T. Liu, J. Li, R. Tao, K.A. Lee, C.E. Siong, *Addressing Gradient Misalignment in Data-Augmented Training for Robust Speech Deepfake Detection*, **IEEE ICASSP 2026**, *GitHub stars: 10+*.
- [7] T.D. Phuong*, **D.-T. Truong***, L.V. Hoang, T.T.T. Nguyen, *Fine-grained Frame Modeling In Multi-head Self-attention For Speech Deepfake Detection*, **IEEE ICASSP 2026**.
- [8] C.Y. Kwok*, **D.-T. Truong***, J.Q. Yip, *Robust Audio Deepfake Detection Using Ensemble Confidence Calibration*, **IEEE ICASSP 2025**.
- [9] **D.-T. Truong**, T.T. Anh, C.E. Siong, *Exploring Speaker Age Estimation on Different Self-Supervised Learning Models*, **IEEE APSIPA ASC 2022**, *GitHub stars: 14*.

- [10] J. Li, K.A. Lee, **D.-T. Truong**, T. Liu, R. Tao, M.W. Mak, *Xi+: Uncertainty Supervision for Robust Speaker Embedding*, **IEEE ICASSP 2026**.
- [11] J.Q Yip, **D.-T. Truong**, D. Ng, C. Zhang, Y. Ma, T.H. Nguyen, C. Ni, S. Zhao, C.E. Siong, B. Ma *ACA-Net: Towards Lightweight Speaker Verification using Asymmetric Cross Attention*, **ISCA INTERSPEECH 2023**.
- [12] T. Gupta, **D.-T. Truong**, T.T. Anh, C.E. Siong, *Estimation of speaker age and height from speech signal using bi-encoder transformer mixture model*, **ISCA INTERSPEECH 2022**, *GitHub stars: 14*.
- [13] H.-T. Luong, **D.-T. Truong**, K.A. Lee, C.E. Siong, *Room impulse responses help attackers to evade deep fake detection*, **IEEE SLT 2024**.
- [14] **D.-T. Truong**, Y. Wang, K.A. Lee, M. Li, H.Nishizaki, C.E. Siong, *A Study of Guided Masking Data Augmentation for Deepfake Speech Detection*, **ASVspoof Workshop 2024**.

PROJECTS

- **Deepfake Speech Detection (Research Project funded by AI Singapore)**
Developed deepfake speech detection methods that enhance both architecture and optimization. Designed temporal-channel modeling for multi-head self-attention [[1]] and addressed gradient misalignment introduced by data augmentation [[6]], improving robustness and cross-condition generalization for synthetic speech detection.
- **Homomorphic Encryption Speaker Verification (Research Project funded by OCBC Bank)**
Implemented an end-to-end speaker verification pipeline, training SOTA models and enabling privacy-preserving inference under fully homomorphic encryption.
- **Interpretability of Speech Signal under Adverse Conditions (Research Project funded by Temasek Lab)**
Developed a novel knowledge distillation-based training method that transfers knowledge from large robust speaker verification models to compact, deployment-friendly models [[2]].
- **Identifying Speaker Characteristics through Audio Profiling (Research Project funded by Temasek Lab)**
Developed a gender-aware mixture-of-experts Transformer model that integrates self-supervised speech representations to accurately estimate speaker characteristics like age and height directly from speech signals [[9]], [[12]].
- **Multi-camera multi-people tracking (Undergraduate Thesis)**
Implemented system a tracking people across multiple non-overlapping cameras using DeepSORT for tracking and ABD-Net model for feature extraction.

HONORS AND AWARDS

- 1st place winner in FAME Challenge, ACM-Multimedia, 2024.
- ISCA INTERSPEECH Travel Grant, 2024.
- NTU Research Scholarship (full-ride scholarship awarded to graduate students).
- KSYS-CUBE Scholarship 2021 (Awarded to 40 excellent HCMUT students annually).
- Acecook Happy Scholarship 2021 (Awarded to 200 outstanding students among 17 universities across Vietnam).
- TRUaire Scholarship 2019 & 2020 (Awarded to 40 selected HCMUT students annually).

SKILLS SUMMARY

- **Languages:** Python, Java, C++, Bash, SQL, LaTeX
- **Frameworks:** HuggingFace, ms-swift, fairseq, PyTorch, Kafka, Faust, MLflow, Vert.x, Django
- **Tools:** Docker, Elasticstack, PostgreSQL, MySQL, Vim, Git

LANGUAGES

- **English:** Fluent (IELTS Academic: 7.5).
- **Vietnamese:** Native.

ACADEMIC SERVICE

Reviewer: ICASSP (2026), INTERSPEECH (2026), SLT (2026).

REFERENCES

- Dr. Chng Eng Siong - Professor @ NTU (Singapore) - ASESChng@ntu.edu.sg - Main supervisor
- Dr. Lee Kong Aik - Professor @ PolyU (Hong Kong) - kongaik.lee@polyu.edu.hk - Co-supervisor
- Dr. Chen Manman - Director @ American Express (Singapore) - manman.chen@aexp.com - Direct Manager
- Dr. Tao Ruijie - AI Researcher @ Zoom (Singapore) - ruijie.tao@u.nus.edu - Mentor
- Dr. Liu Tianchi - Senior Researcher @ Tencent (Singapore) - tianchi.liu@u.nus.edu - Mentor